Adaptive Reasoning Agent

Query

Observation

MLLM

Reasoning

Action

Answer

Tool Chain

propose

Global-Level Retrieval

Segment-Level Retrieval

Frame-Level Retrieval

Hierarchical Construction

Verify

Control Stategy

Frame Inspect

Scene Graph Representation

Global-level Graph

Segment-level Graph

Frame-level Graph

Hierarchical Graph Database

Caption Database

Long Video Input

